

Module 5 Summary



Module 5 Review, Reflect, Consolidate, and Practice

Module 5 focused on futures contracts, options contracts, hedging, and financial risk management. Throughout the module, you learned how derivative markets help firms reduce price uncertainty, stabilize cash flows, and manage financial exposure. You also examined the trade-offs between certainty, flexibility, liquidity, and opportunity cost in hedging decisions.

This section summarises what you have learned and links it explicitly to the course learning outcomes.

What You Learned in Module 5

Why Derivatives Markets Exist

You learned that businesses face constant exposure to price risk from:

- Commodity prices
- Exchange rates
- Interest rates
- Input costs
- Output prices

You learned that:

- Price uncertainty creates major planning and budgeting problems.
- Futures and options contracts allow firms to reduce uncertainty.
- Hedging focuses on stabilizing revenues and costs rather than maximizing speculative profit.

This directly supports:

CLO1: Explain key concepts and theoretical foundations of financial analytics and their role in strategic decision-making.

You can now explain the economic role of derivatives markets and why firms hedge risk.

Futures Contracts and Hedging

You learned that a futures contract is a standardized agreement to buy or sell an asset at a predetermined future price.

You learned the main features of futures markets, including:

- Standardization
- Clearinghouses
- Margin accounts
- Marking-to-market
- Daily settlement

You also learned that:

- Futures reduce uncertainty but eliminate upside potential.
- Margin calls can create liquidity pressure.
- Clearinghouses reduce counterparty risk and improve market stability.

You now understand that:

- Futures convert volatile prices into predictable outcomes.
- Hedging involves trade-offs between certainty and opportunity.

This supports:

CLO2: Apply quantitative and time-series techniques to model risk, return, and financial performance in real-world contexts.

You can now calculate futures gains and losses and interpret hedging outcomes quantitatively.

Options Contracts and Financial Flexibility

You learned that options provide a right, but not an obligation, to buy or sell an asset.

You distinguished between:

- Call options
- Put options

You learned that:

- Option buyers have limited downside risk.

- Option sellers face potentially very large losses.
- Options preserve upside potential while providing downside protection.

You also learned key concepts including:

- Strike price
- Premium
- Intrinsic value
- Time value
- Moneyness
- Expiration

This supports:

CLO2: Apply quantitative and time-series techniques to model risk, return, and financial performance in real-world contexts.

You can now calculate option payoffs and interpret protective hedging strategies.

Option Pricing and Volatility

You learned that option prices depend heavily on:

- Spot price
- Strike price
- Time to expiry
- Volatility

You were introduced to important option sensitivities, including:

- Delta
- Theta
- Vega

You learned that:

- Higher volatility increases option premiums.
- More time increases option value.
- Options respond asymmetrically to market movements.

This supports:

CLO2: Apply quantitative and time-series techniques to model risk, return, and financial performance in real-world contexts.

You can now interpret how changing market conditions affect derivative pricing.

Futures vs Options

Throughout the module, you compared futures and options as risk-management tools.

You learned that:

- Futures provide stronger price certainty.
- Options preserve upside potential but require premium payments.
- Futures expose traders to margin calls.
- Options provide insurance-like protection.

You now understand that:

- Futures are often preferred when certainty is critical.
- Options are often preferred when flexibility and upside participation matter.

This supports:

CLO3: Use business analytics tools to develop, evaluate, and communicate predictive insights that support financial problem-solving.

You can now evaluate different hedging strategies using financial reasoning.

Communicating Financial Risk Clearly

Throughout the module, you interpreted and communicated:

- Hedging strategies
- Futures payoffs
- Option payoffs
- Risk exposures
- Hedging trade-offs

You learned how to:

- Explain derivatives in business language.
- Compare risk-management strategies clearly.
- Connect analytical outputs to financial decision-making.

This supports:

CLO4: Create and communicate financial insights using effective visual, written, and oral formats suited to both technical and non-technical stakeholders.

You can now explain derivatives and hedging strategies to both technical and non-technical audiences.

Practice Tasks

Practice Task 1: Futures Hedging

A wheat producer expects to sell 5,000 tonnes of wheat in three months and fears falling prices.

Current futures price = \$620 per tonne.

Explain:

- Why the producer may choose to short futures contracts
 - What happens if prices fall sharply
 - What opportunity cost arises if prices rise instead
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Practice Task 2: Margin Calls

A trader opens a futures position requiring:

- Initial margin = \$15,000
- Maintenance margin = \$11,000

After market losses, the margin balance falls to \$9,500.

Calculate:

- The margin call amount

Explain:

- Why margin calls are important for market stability
 - Why liquidity risk matters for hedgers
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Practice Task 3: Corn Call Option

A corn call option has:

- Spot price = \$450
- Strike price = \$460
- Premium = \$15

Calculate:

- The breakeven price

Interpret:

- When the option becomes profitable
 - The maximum loss
 - Why upside gain is potentially unlimited
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Practice Task 4: Coffee Put Option

A coffee put option has:

- Spot price = \$4,200 per tonne
- Strike price = \$4,300
- Premium = \$120

The market price falls to \$4,000.

Calculate:

- The put payoff
- The net gain per tonne

Explain:

- Why put options provide downside protection
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Practice Task 5: Gold Producer Protective Put

A gold producer buys put options with:

- Strike price = \$3,000 per ounce
- Premium = \$80 per ounce
- Exposure = 5,000 ounces

The gold price falls to \$2,700.

Calculate:

- The put gain
- The premium cost
- The net protection amount
- The effective price floor

Explain:

- Why protective puts resemble insurance
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Answers available [here \(https://learn.usc.edu.au/courses/35359/files/3459900?wrap=1\)](https://learn.usc.edu.au/courses/35359/files/3459900?wrap=1).

Reflect

- Why are options described as asymmetric financial instruments?
- What is the difference between intrinsic value and time value?
- Why does higher volatility increase option premiums?

- When might a firm prefer options over futures despite higher upfront cost?
 - Why is selling options potentially much riskier than buying them?
 - Why can a successful hedge still create liquidity stress?
 - Why do firms hedge rather than speculate?
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Workshop

Tasks (<https://learn.usc.edu.au/courses/35359/files/3459860?wrap=1>)